INCH-POUND

MIL-M-28787/305A(EC)
AMENDMENT 3
7 September 1994
SUPERSEDING
AMENDMENT 2
14 January 1992

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

CONTROLLER, INTERRUPT, DIGITAL, KEY CODE ZJX

This amendment forms a part of Military Specification MIL-M-28787/305A(EC), dated 14 May 1982, and is approved for use by the Crane Division, Naval Surface Warfare Center, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGES 4, 5, 6, 7 AND 8

*TABLE I: Delete and substitute new table I as printed on pages 5, 6, 7 and 8 of this amendment.

PAGE 9

4.4.2: Delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through BB shall be performed in any order with the following exceptions. Test conditions A, B, C, D and E are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, C, D or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D and E where applicable, followed by the remaining tests in the sequence."

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4.5.1: Delete and substitute the following:

"4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions F through BB in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions F through BB shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2."

4.5.3: Delete and substitute the following:

"4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests F through X and BB shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests F through BB shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I."

PAGE 10

6.4, Module description, second line: Delete "100°C" and substitute "85°C".

*6.4, Module description, Input current, Low level: Add the following line:

$\overline{IE}$	0.0 min, -1.2 max	0.0 min, -1.2 max	mA

PAGE 12

*FIGURE 2: Delete and substitute new figure 2 as printed on page 9 of this amendment.

PAGE 29

*TABLE VIII, Test condition G14, Max limit: Delete "-0.87" and substitute "-1.2".

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PAGE 44

TABLE XV, footnote 1, second line: Delete "GE" (two places), "GAR" (two places), "IE", and "ID" and substitute "$\overline{GE}$" (two places), "$\overline{GAR}$" (two places), "$\overline{IE}$" and "$\overline{ID}$", respectively.

TABLE XV, footnote 10, first line: Delete "GE", "IE", and "GAR" and substitute "$\overline{GE}$", "$\overline{IE}$", and "$\overline{GAR}$", respectively.

PAGE 53

TABLE XVIII, Test conditions S3 and S5, Parameter column: Delete "$\overline{ID}$" and "ID" and substitute "$\overline{ID}$" and "$\overline{ID}$", respectively.

PAGE 59

TABLE XX: Delete and substitute new table XX as printed on page 10 of this amendment.

PAGE 60

TABLE XXI: Delete and substitute new table XXI as printed on page 11 of this amendment.

PAGE 61

*FIGURE 24: Delete and substitute new figure 24 as printed on page 12 of this amendment.

PAGE 70

TABLE XXVII, Test conditions BB1 through BB16, Parameter column: Delete "IE" (sixteen places) and substitute "$\overline{IE}$" (sixteen places).

PAGE 71

TABLE XXVII, Test conditions BB17 and BB32, Parameter column: Delete "IE" (two places) and substitute "$\overline{IE}$" (two places).

PAGES 72 AND 73

TABLE XXVII, Test conditions BB33 through BB56, Parameter column: Delete "IE" (twenty-four places) and substitute "$\overline{IE}$" (twenty-four places).

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NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Preparing activity:
Navy - NW

(Project 5963-N867)

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TABLE I. Module electrical requirements.

Test cond	Parameter	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation Initial	II	S	10	INF	10	INF	10	INF	10	INF	10	INF	MOhm
	End-of-life	II	S	1	INF	1	INF	1	INF	1	INF	1	INF	MOhm
B	Voltage (Vcc) stress test	III	S	----------		----------		Must pass remaining tests		----------		----------		
C1-C29	Input breakdown current	IV	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
D1-D18	Input clamp diode voltage	V	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
E1-E19	Output short circuit current	VI	S	-40	-100	-40	-100	-40	-100	-40	-100	-40	-100	mA
F1-F14	Input current, high level	VII	X	0	20	0	20	0	20	0	20	0	20	µA
F15-F20	Input current, high level	VII	X	0	40	0	40	0	40	0	40	0	40	µA
F21-F29	Input current, high level	VII	X	0	60	0	60	0	60	0	60	0	60	µA
* G1-G13	Input current, low level	VIII	X	0.0	-0.87	0.0	-0.87	0.0	-0.87	0.0	-0.87	0.0	-0.87	mA
* G14	Input current, low level	VIII	X	0.0	-1.2	0.0	-1.2	0.0	-1.2	0.0	-1.2	0.0	-1.2	mA
* G15, G16	Input current, low level	VIII	X	0.0	-0.87	0.0	-0.87	0.0	-0.87	0.0	-0.87	0.0	-0.87	mA
G17	Input current, low level	VIII	X	0.0	-0.433	0.0	-0.433	0.0	-0.433	0.0	-0.433	0.0	-0.433	mA
G18	Input current, low level	VIII	X	0.0	-2.17	0.0	-2.17	0.0	-2.17	0.0	-2.17	0.0	-2.17	mA
G19-G21	Input current, low level	VIII	X	0.0	-0.108	0.0	-0.108	0.0	-0.108	0.0	-0.108	0.0	-0.108	mA
G22-G29	Input current, low level	VIII	X	0.0	-0.163	0.0	-0.163	0.0	-0.163	0.0	-0.163	0.0	-0.163	mA
H1, H2, H4-H20	Output voltage, high level	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
H3	Output voltage, high level	IX	X	2.4	4.55	2.4	4.55	2.4	4.55	2.4	4.55	2.4	4.55	V
J1-J20	Output voltage, low level	X	X	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	V
K1-K3	High-impedance leakage current, high level	XI	X	-1	50	-1	50	-1	50	-1	50	-1	50	µA
K4-K11	High-impedance leakage current, high level	XI	X	-1	150	-1	150	-1	150	-1	150	-1	150	µA
K12-K14	High-impedance leakage current, high level	XI	X	-1	100	-1	100	-1	100	-1	100	-1	100	µA
L1-L3	High-impedance leakage current, low level	XII	X	1	-50	1	-50	1	-50	1	-50	1	-50	µA
L4-L11	High-impedance leakage current, low level	XII	X	1	-150	1	-150	1	-150	1	-150	1	-150	µA
L12-L14	High-impedance leakage current, low level	XII	X	1	-100	1	-100	1	-100	1	-100	1	-100	µA
M	Output leakage current	XIII	X	0	250	0	250	0	250	0	250	0	250	µA
N	Power supply current	XIV	X	26	260	27	275	27	275	30	310	30	310	mA

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
P1-P6	Dynamic functional test	XV	X	See table XV										
Q1-Q5	Transition time (tTLH) CL = 50 pF	XVI	S	1	17	1	17	1	22	1	17	1	27	nS
R1-R6	Transition time (tTHL) CL = 50 pF	XVII	S	1	6	1	6	2	11	1	6	2	16	nS
S1, S2	Propagation delay time (tPLH), CL = 50pF	XVIII	S	4	47	4	47	5	54	4	47	6	59	nS
S3	Propagation delay time (tPLH), CL = 50pF	XVIII	S	4	42	4	42	5	51	4	42	6	57	nS
S4	Propagation delay time (tPLH), CL = 50pF	XVIII	S	2	13	2	13	2	14	2	13	2	15	nS
S5	Propagation delay time (tPLH), CL = 50pF	XVIII	S	2	23	2	23	2	28	2	23	3	31	nS
S6	Propagation delay time (tPLH), CL = 50pF	XVIII	S	8	82	8	82	9	93	8	82	10	101	nS
S7, S8	Propagation delay time (tPLH), CL = 50pF	XVIII	S	5	57	5	57	6	66	5	57	7	72	nS
S9	Propagation delay time (tPLH), CL = 50pF	XVIII	S	6	66	6	66	7	75	6	66	8	81	nS
S10	Propagation delay time (tPLH), CL = 50pF	XVIII	S	4	42	4	42	4	48	4	42	5	52	nS
S11	Propagation delay time (tPLH), CL = 50pF	XVIII	S	4	45	4	45	5	51	4	45	5	55	nS
S12	Propagation delay time (tPLH), CL = 50pF	XVIII	S	3	32	3	32	3	38	3	32	4	42	nS
S13	Propagation delay time (tPLH), CL = 50pF	XVIII	S	5	56	5	56	6	61	5	56	6	64	nS
S14	Propagation delay time (tPLH), CL = 50pF	XVIII	S	2	25	2	25	3	30	2	25	3	33	nS
S15	Propagation delay time (tPLH), CL = 50pF	XVIII	S	8	82	8	82	9	93	8	82	10	101	nS
S16, S17	Propagation delay time (tPLH), CL = 50pF	XVIII	S	5	57	5	57	6	66	5	57	7	72	nS
S18	Propagation delay time (tPLH), CL = 50pF	XVIII	S	7	74	7	74	8	85	7	74	9	92	nS
T1, T2	Propagation delay time (tPHL), CL = 50pF	XIX	S	4	47	4	47	5	54	4	47	6	59	nS
T3	Propagation delay time (tPHL), CL = 50pF	XIX	S	4	42	4	42	5	51	4	42	6	57	nS
T4	Propagation delay time (tPHL), CL = 50pF	XIX	S	2	13	2	13	2	14	2	13	2	15	nS
T5	Propagation delay time (tPHL), CL = 50pF	XIX	S	2	23	2	23	2	28	2	23	3	31	nS

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
T6	Propagation delay time (tPHL), CL = 50pF	XIX	S	8	82	8	82	9	93	8	82	10	101	nS
T7, T8	Propagation delay time (tPHL), CL = 50pF	XIX	S	5	57	5	57	6	66	5	57	7	72	nS
T9	Propagation delay time (tPHL), CL = 50pF	XIX	S	6	66	6	66	7	75	6	66	8	81	nS
T10	Propagation delay time (tPHL), CL = 50pF	XIX	S	4	42	4	42	4	48	4	42	5	52	nS
T11	Propagation delay time (tPHL), CL = 50pF	XIX	S	4	45	4	45	5	51	4	45	5	55	nS
T12	Propagation delay time (tPHL), CL = 50pF	XIX	S	3	32	3	32	3	38	3	32	4	42	nS
T13	Propagation delay time (tPHL), CL = 50pF	XIX	S	5	56	5	56	6	61	5	56	6	64	nS
T14	Propagation delay time (tPHL), CL = 50pF	XIX	S	2	30	2	30	3	30	2	30	3	33	nS
T15	Propagation delay time (tPHL), CL = 50pF	XIX	S	8	82	8	82	9	93	8	82	10	101	nS
T16, T17	Propagation delay time (tPHL), CL = 50pF	XIX	S	5	57	5	57	6	66	5	57	7	72	nS
T18	Propagation delay time (tPHL), CL = 50pF	XIX	S	7	74	7	74	8	85	7	74	9	92	nS
* U1-U20	Output crosstalk, low level	XX	Q	0	500	0	500	0	500	0	500	0	500	mV
* V1-V20	Output crosstalk, high level	XXI	Q	0	500	0	500	0	500	0	500	0	500	mV
W	Power supply transient amplitude	XXII	Q	0	400	0	400	0	400	0	400	0	400	mV
X	Power supply filter capacitance	XXIII	S	3.15	5.85	3.15	5.85	3.15	5.85	3.15	5.85	3.15	5.85	µF
Y1-Y13, Y15, Y16	Input capacitance	XXIV	Q	2	25	2	25	2	25	2	25	2	25	pF
Y14	Input capacitance	XXIV	Q	2	45	2	45	2	45	2	45	2	45	pF
Y17, Y18	Input capacitance	XXIV	Q	2	35	2	35	2	35	2	35	2	35	pF
Z1-Z8	Input/output capacitance	XXV	Q	2	50	2	50	2	50	2	50	2	50	pF
Z9-Z11	Input/output capacitance	XXV	Q	2	40	2	40	2	40	2	40	2	40	pF
AA1	Output capacitance	XXVI	Q	2	35	2	35	2	35	2	35	2	35	pF
AA2-AA4	Output capacitance	XXVI	Q	2	25	2	25	2	25	2	25	2	25	pF
AA5-AA9	Output capacitance	XXVI	Q	2	40	2	40	2	40	2	40	2	40	pF

See footnotes at end of table.

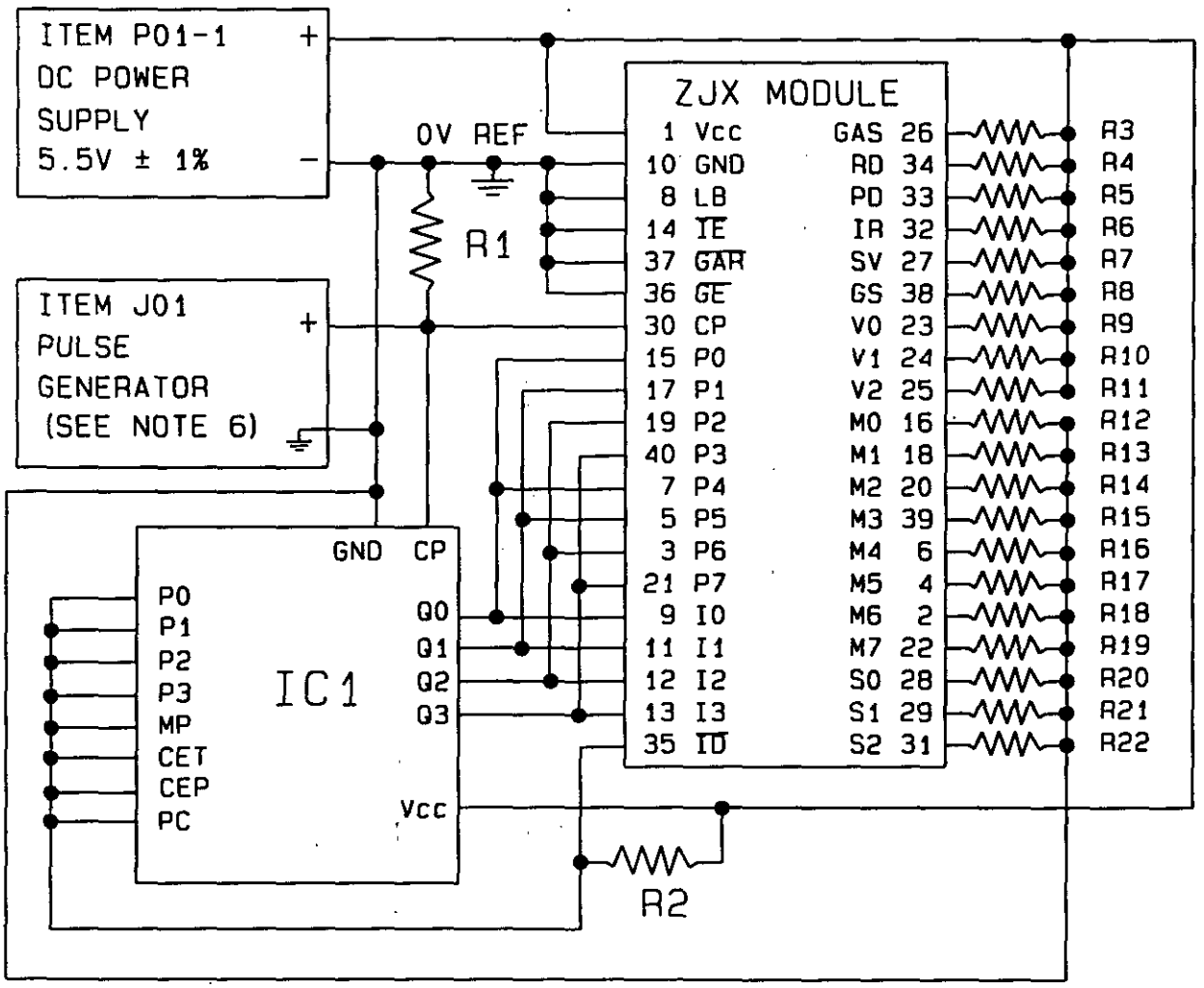
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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/ - 3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
BB1-BB16	Disable/enable recovery time (tPLZ, tPZL), CL = 50 pF	XXVII	S	5	52	5	52	5	58	5	52	6	62	nS
BB18,BB20, BB22,BB24, BB26,BB28, BB30,BB32	Disable/enable recovery time (tPZH), CL = 50 pF	XXVII	S	5	57	5	57	5	63	5	57	6	67	nS
BB17,BB19, BB21,BB23, BB25,BB27, BB29,BB31	Disable/enable recovery time (tPHZ), CL = 50 pF	XXVII	S	6	70	6	70	7	76	6	70	7	80	nS
BB33-BB38, BB40,BB42, BB44-BB50, BB52,BB54, BB56	Disable/enable recovery time (tPLZ, tPZL, tPZH), CL = 50 pF	XXVII	S	5	59	5	59	6	65	5	59	7	69	nS
BB39,BB41, BB43,BB51, BB53,BB55	Disable/enable recovery time (tPHZ), CL = 50 pF	XXVII	S	6	65	6	65	7	71	6	65	7	75	nS

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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NOTES:
1. IC1 = 9316 or equivalent.
2. R1 = 51 Ohms ± 5 percent, 1/2 W.
3. R2, R12-R22 = 1.2 kOhms ± 5 percent, 1/4 W.
4. R3-R11 = 420 Ohms ± 5 percent, 1/4 W.

*FIGURE 2. Operating temperature, component temperature, life test circuit and waveforms.

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TABLE XX. Output crosstalk, low level test.

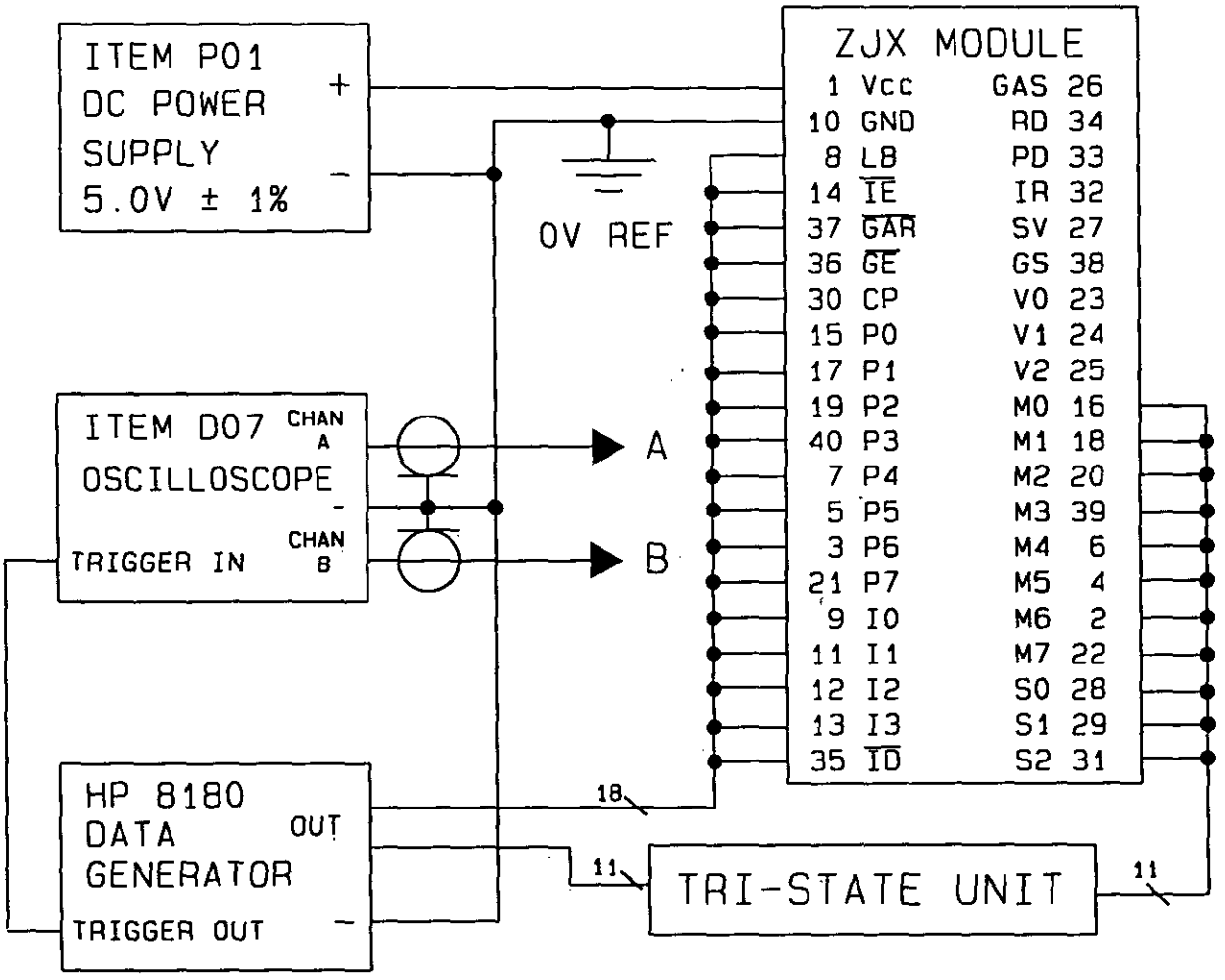
Test cond	Signal name	Pin no.	Initial limits 25 ± 5°C		Units	Notes	Figs.
			Min	Max			
U1	V0	23	0	500	mV	1-4	23, 24
U2	V1	24					
U3	V2	25					
U4	S0	28					
U5	S1	29					
U6	S2	31					
U7	M0	16					
U8	M1	18					
U9	M2	20					
U10	M3	39					
U11	M4	6					
U12	M5	4					
U13	M6	2					
U14	M7	22					
U15	GAS	26					
U16	RD	34					
U17	PD	33					
U18	IR	32					
U19	SV	27					
U20	GS	38	0	500	mV	1-4	23, 24

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TABLE XXI. Output crosstalk, high level test.

Test cond	Signal name	Pin no.	Initial limits 25 ± 5°C		Units	Notes	Figs.
			Min	Max			
V1	V0	23	0	500	mV	1-4	23, 24
V2	V1	24					
V3	V2	25					
V4	S0	28					
V5	S1	29					
V6	S2	31					
V7	M0	16					
V8	M1	18					
V9	M2	20					
V10	M3	39					
V11	M4	6					
V12	M5	4					
V13	M6	2					
V14	M7	22					
V15	GAS	26					
V16	RD	34					
V17	PD	33					
V18	IR	32					
V19	SV	27					
V20	GS	38	0	500	mV	1-4	23, 24

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NOTES:
1. During changes to instruction codes I0-I3, clock must be high.
2. Load all outputs per figure 12, including M0-M7 and S0-S2.

*FIGURE 24. Output crosstalk, high and low level, test circuit for tables XX and XXI.

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